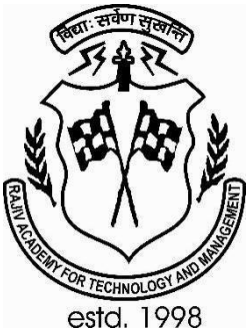
RAJEEV ACADEMY FOR TECHNOLOGY & MANAGEMENT, (MATHURA)



MAJOR-PROJECT REPORT

On

LIBRARY MANAGEMENT SYSTEM

Submitted for the partial fulfillment towards the award of the degree in

MASTER OF COMPUTER APPLICATION

Of

Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow

Submitted By

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Roll Number: 2401730140021
2025-26

Under the Guidance of

MR . MAYANK SHAKKERWAL
Department of MCA Session:

PREFACE

A Library Management System plays a vital role in organizing, maintaining, and managing library resources efficiently in the modern digital era. With the rapid growth of information and the increasing number of users, traditional manual methods of library management have become time-consuming, error-prone, and inefficient. This project aims to develop a Library Management System that automates routine library activities and provides a reliable, user-friendly platform for both librarians and users.

The Library Management System is designed to handle essential operations such as book cataloging, member registration, book issue and return, fine calculation, and report generation. By integrating these functions into a single system, the project reduces paperwork, minimizes human errors, and ensures quick access to accurate information. The system enhances productivity and helps librarians manage resources effectively while improving the overall user experience.

This project also focuses on data security, accuracy, and ease of use. It provides controlled access to users based on their roles and ensures that records are stored safely for future reference. The Library Management System serves as an efficient solution for academic institutions, public libraries, and organizations seeking to modernize their library operations and improve service quality.

In conclusion, the Library Management System represents a step toward digital transformation in library services, offering a scalable, reliable, and efficient approach to managing knowledge resources in an organized and systematic manner.

.

CERTIFICATE OF GUIDANCE

To whom it may Concern

Department Of Computer Science

This is certified that the project entitles "LIBRARY MANAGEMENT SYSTEM" has been developed under the guidance and supervision during session 2025-26 by following student of M.C.A IV Semester.

Vishal kumar

It is further certified that these students have submitted the present volume as a project report to the department of computer science. This is for partial fulfillment of award of degree of Master of Computer Application from Dr. APJ Abdul Kalam Technical University, Lucknow.

MR . MAYANK SHAKKERWAL

(Project Guide)

STUDENT'S DECLARATION

I, **VISHAL KUMAR**, hereby declare that this work entitled "**LIBRARY MANAGEMENT SYSTEM"** is the result of summer training under taken. The findings and conclusions expressed in this report are genuine, authentic and are for academic purpose. Any resemblance to earlier research work is purely coincidental.

VISHAL KUMAR
(MCA IV Sem)
Roll No. 2401730140021

REPORT OF SUPERVISIOR CERTIFICATION

I, **MR . MAYANK SHAKKERWAL,** hereby certify that this research work entitled "**LIBRARY MANAGEMENT SYSTEM"** is the result of research under taken by **VISHAL KUMAR (2401730140021)** the findings and conclusions expressed in this report are genuine, authentic and are for academic purpose only. Any resemblance to earlier work is purely coincidental.

Signature

MR . MAYANK SHAKKERWAL

ABSTRACT

The Library Management System is a comprehensive software application designed to automate, manage, and optimize the day-to-day operations of a library. With the increasing volume of information and growing number of library users, traditional manual systems have become inefficient, time-consuming, and prone to errors. This project aims to provide a digital solution that simplifies library processes, improves accuracy, and enhances overall efficiency.

The system facilitates various core functions such as book cataloging, classification, member registration, book issue and return management, reservation handling, fine calculation, and report generation. It maintains a centralized database that stores complete and up-to-date information about library resources and users, enabling quick data retrieval and efficient tracking of books. The system ensures that librarians can easily monitor book availability, overdue items, and user activity.

In addition, the Library Management System incorporates role-based access control to ensure data security and integrity. Librarians and administrators are granted appropriate privileges to manage records, while users can search for books, check availability, and view their borrowing history. Automation of routine tasks significantly reduces paperwork, minimizes human errors, and saves time for both library staff and users.

The proposed system is designed to be scalable, reliable, and user-friendly, making it suitable for academic institutions, public libraries, and corporate organizations. By adopting this Library Management System, libraries can modernize their operations, improve service quality, and provide a more organized and efficient approach to knowledge management in the digital age.

ACKNOWLEDGEMENT

At the onset I must bow down in reverence to the almighty that blessed us with the understanding & prevalence that is needed in this kind of project report.

With great pleasure I express my heartiest thanks to **MR . MAYANK SHAKKERWAL** (Head – MCA Deptt.). I would like to extend my sincere thanks .of Computer Science Studies], without whose unrelated support and guidance, this project would just not have been possible. I am very thankful for his invaluable guidance, support, and affable & friendly nature. He/She guided me at each and every stage of project.

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At last I would like to extend my sincere thanks to all the respondents to whom I visited for giving their support and valuable information, which helps me in completing my project work.

VISHAL KUMAR

MCA IV Sem

Roll No.

2401730140021

TABLE OF CONTENTS

PREFACE.....i

CERTIFICATE OF GUIDANCE..... ii

STUDENT'S DECLARATION.....iii

REPORT OF SUPERVISIOR CERTIFICATON.....iv

ABSTRACT..... v

ACKNOWLEDGEMENT..... vi

TABLE OF CONTENT……vii--vii

CHAPTER-1.....1

1. INTRODUCTION.....2

1.1 INTRODUCTION OF THE PROJECT..... 2
1.2 PROBLEM STATEMENT..... 2
1.3 OBJECTIVE OF THE PROJECT.....3
1.4 FEATURES OF THE PROJECT..... 4
1.5 SCOPE OF THE PROJECT..... 5
CHAPTER-2.....6

2. FEASIBILITY STUDY.....7

2.1 TECHNICAL FEASIBILITY..... 7
2.2 ECONOMIC FEASIBILITY..... 8
2.3 OPERATIONAL FEASIBILITY..... 8
2.4 ADDITIONAL FEASIBILITY..... 9
2.5 SUMMARY AND FINAL ASSESSMENT.....9
CHAPTER-3..... 10

3. SOFTWARE REQUIREMENT SPECIFICATION..... 12

3.1 INTRODUCTION TO SRS..... 12
3.2 SOFTWARE REQUIREMENTS.....12
3.3 HARDWARE REQUIREMENTS.....13
CHAPTER-4.....13

4. SYSTEM DESIN..... 15

4.1 PROCESS FLOW DIAGRAM..... 15
4.2 ER DIAGRAM..... 15

4.3 DATA FLOW DIAGRAM..... 17
CHAPTER-5.....18

5. TOOLS AND TECHNOLOGY..... 19

5.1 DEVELOPMENT TOOLS.....19
5.2 TECHNOLOGY USED.....20
5.3 WEB-SERVER.....21
CHAPTER-6.....22
6. SCREENSHOTS..... 23

7. CODING..... 29

confiq.php..... 29
Index.php.....30
registration.php..... 32
login.php..... 35
Profile.php.....38
notification.php.....42
Book.php.....44
varification.php..... 46
CHAPTER-8.....48

8. TESTING..... 49

8.1 WHAT IS TESTING.....49
8.2 TYPES OF TESTING.....49
8.3 TESTCASES OF THE PROJECT.....51

9.CONCLUSION..... 54

REFERENCES.....…

TABLE OF FIGURES
FLOWCHART

FIGURE 1 PROCESS FLOW DIAGRAM.....14
FIGURE 2 COMPONENTS OF AN ER DIAGRAM..... 15
FIGURE 3 ER DIAGRAM..... 16
FIGURE 4 SYMBOLS FOR DFD.....17
FIGURE 5 PROFILE.PHP.....23
FIGURE 6 DASHBOARD.....23
FIGURE 7 LOGIN.PHP.....24
FIGURE 8 REGISTRATION FORM.....24
FIGURE 9 STATUS.PHP.....25
FIGURE 10 FINE.PHP..... 25
FIGURE 11 ADDBOOK..... 26
FIGURE 12 DATABASE 1.....26
FIGURE 13 DATABASE 2..... 27

CHAPTER-1

1. INTRODUCTION

1.1 INTRODUCTION OF THE PROJECT

Library Management System project is developed to provide an efficient and automated solution for managing library operations in a systematic and organized manner. In today's digital age, libraries handle a large volume of books, journals, and users, making manual record-keeping methods inadequate and error-prone. This project aims to replace the traditional manual library system with a computerized system that simplifies and streamlines day-to-day library activities.

The main purpose of this project is to automate key library functions such as book cataloging, member registration, book issue and return, fine calculation, and report generation. By maintaining all records in a centralized database, the system ensures accurate data storage, quick retrieval of information, and easy updating of records. This reduces paperwork, minimizes human errors, and saves valuable time for both librarians and users.

The Library Management System project is designed to be user-friendly and secure. It provides role-based access to different users such as administrators, librarians, and members. Administrators can manage book inventories and user records, while members can search for books, check availability, and view their borrowing history. This structured access ensures data security and integrity.

Overall, this project focuses on improving the efficiency, reliability, and quality of library services. By implementing the Library Management System, libraries can modernize their operations, enhance user satisfaction, and ensure effective management of knowledge resources in a fast-growing information environment.

1.2 PROBLEM STATEMENT

Libraries that rely on traditional manual systems face several challenges in managing their daily operations efficiently. Maintaining records of books, members, and transactions using registers or paper files is time-consuming and prone to errors. As the number of books and users increases, it becomes difficult to track issued and returned books, monitor due dates, and calculate fines accurately. Searching for specific books or user details also requires significant time and effort.

Manual systems often lead to problems such as data redundancy, loss or damage of records, incorrect entries, and lack of real-time information about book availability. These issues result in delays, reduced productivity, and poor user satisfaction. Additionally, ensuring data security and maintaining accurate records becomes challenging without proper access control mechanisms.

Therefore, there is a need for an automated Library Management System that can efficiently manage library resources, reduce human effort, improve accuracy, and provide quick access to information. The proposed system aims to address these problems by offering a centralized, secure, and user-friendly platform for managing all library-related activities effectively.

1.3 OBJECTIVE OF THE PROJECT

The main objective of the Library Management System project is to design and develop an automated system that efficiently manages library operations and replaces the traditional manual methods. The project aims to simplify library activities, improve accuracy, and enhance the overall efficiency of library services.

The specific objectives of the project are as follows:

1. To automate routine library operations such as book issue, return, renewal, and fine calculation.
2. To maintain accurate and up-to-date records of books, members, and transactions in a centralized database.
3. To reduce paperwork and minimize human errors associated with manual record-keeping.
4. To provide quick and easy access to information such as book availability and user details.
5. To implement role-based access control to ensure data security and integrity.
6. To generate reports related to book inventory, issued books, overdue items, and user activities.
7. To improve the efficiency and productivity of library staff.
8. To enhance user satisfaction by providing a reliable and user-friendly system.

Overall, the objective of this project is to develop a scalable, secure, and efficient Library Management System that supports effective management of library resources and services in a modern digital environment.

1.4 FEATURES OF THE PROJECT

The **Library Management System** includes a set of useful features designed to automate and streamline library operations, making the process more efficient, accurate, and user-friendly. The key features of the project are:

1. **User-Friendly Interface**
   A simple and intuitive interface that makes it easy for librarians and users to interact with the system

2. **Book Management**
   Add, update, delete, and manage book records including details such as title, author, publication, category, and availability.

3. **Member Management**
   Register new members, update member profiles, and manage member information such as contact details and library ID.

4. **Book Issue and Return**
   Efficient handling of book issuance and returns with accurate recording of dates, member details, and status.

5. **Fine Calculation**
   Automatic calculation of fines for overdue books based on predefined rules or time limits.

6. **Search Functionality**
   Quick search options to find books by title, author, category, or ISBN to save time and improve efficiency.

7. **Real-Time Availability Status**
   Displays real-time book availability so users can instantly check if a book is available or issued.

8. **Report Generation**
   Generate detailed reports such as issued books list, overdue books, member activity logs, inventory reports, etc.

9. **Centralized Database**
   All data is stored in a centralized system, ensuring consistency, easy access, and secure storage of records.

10. **Role-Based Access Control**
    Different access levels for administrators, librarians, and users to ensure security and prevent unauthorized actions.

11. **Data Backup and Recovery**
    Supports data backup to prevent loss of important records and enables easy recovery when necessary.

12. **Efficient Record Keeping**
    Maintains accurate and systematic records of every transaction, eliminating manual errors.

1.5 SCOPE OF THE PROJECT

The *scope of a project* refers to what the project will cover and what it will deliver — in other words, the **boundaries and extent of the work to be done** to complete the project successfully. It defines the features, functions, tasks, and goals that the system must include to fulfill the expectations of users and stakeholders. A well- defined scope helps keep the project focused, prevents unnecessary features from being added, and ensures that the system meets its intended purposes.

In the case of the **Library Management System project**, the scope describes the key functionalities and areas where the system will provide support:

Automation of Library Work
The system automates routine library tasks such as recording book issues and returns, updating inventory counts, managing member data, and computing overdue fines. This means the system replaces manual entry with reliable computerized processes.

Book and Inventory Management
The scope includes managing book records — storing details like title, author, publisher, category, and current availability — so that librarians can easily add, update, or search for books.

User/Member Services
The system tracks member registrations and provides a platform where members can search for books, check book availability, and view their borrowing history.

Circulation Tracking
The system maintains records of which member has borrowed which book and when it is due to be returned. This prevents lost records and helps reduce overdue returns.

CHAPTER-2

2. FEASIBILITY STUDY – LIBRARY MANAGEMENT SYSTEM

A feasibility study is conducted to determine whether a proposed project is practical, achievable, and worthwhile before investing significant time and resources into its development. A feasibility study objectively evaluates the technical, economic, and operational aspects of the proposed Library Management System to ensure that the project is viable and can be successfully implemented.

2.1 Technical Feasibility

Technical feasibility assesses whether the necessary technology, resources, hardware, and software are available to develop and support the Library Management System. Key questions addressed include:

Availability of Technology:
The required software tools, programming languages, database systems, and development environments are readily available for use in system development.

Hardware Infrastructure:
The current computer systems, servers, and networking infrastructure in the organization are capable of supporting the system without significant upgrades.

Technical Expertise:
The development team has the required technical skills, knowledge, and experience to design, implement, and maintain the system.

System Integration:
The proposed system can be integrated with existing academic or student information systems, if required, to share data such as member IDs and user authentication.

Maintenance and Support:
The system can be maintained with existing tools and processes, and future upgrades and enhancements can be accommodated without using unsupported technology.

Conclusion:
Based on the evaluation of technical resources, infrastructure, and expertise, the Library Management System is technically feasible and can be developed using available technology and equipment at a reasonable cost.

2.2 Economic Feasibility

Economic feasibility, also known as cost- benefit analysis, determines whether the project is financially viable. It examines costs and benefits to assess whether expected benefits outweigh the expenses.

Development Costs:
Costs include software development, customization, user interface design, database setup, coding, and system testing.

Hardware and Software Costs:
Minimal hardware costs may be incurred if the current systems require upgrades. Software costs include database licenses, server software, and development tools where required.

Training Costs:
Librarians, administrators, and end- users may require training on how to use the new system effectively.

Maintenance Costs:
Costs for ongoing support, periodic upgrades, bug fixes, and technical assistance.

Cost Savings:
The system reduces manual record- keeping, minimizes errors, lowers labor costs for repetitive tasks, and reduces paper usage.

Return on Investment (ROI):
In the long term, the system will save time and resources, leading to higher efficiency and cost reduction compared to continued manual operations.

Conclusion:
After analyzing all expenses and savings from improved productivity, the proposed Library Management System is economically feasible as the long- term benefits outweigh the costs.

2.3 Operational Feasibility

Operational feasibility focuses on whether the system will function effectively within the existing operational environment and whether users will accept it.

User Acceptance:
The system is designed with user- friendly interfaces that allow librarians and members to interact easily with minimal training.

Usability:
Essential functions like book search, issuance, return processing, and availability updates are intuitive and streamline daily activities.

Improvement Over Manual System:
The automated system significantly improves accuracy, efficiency, and reliability compared to manual logbooks and files.

Compatibility with Organizational Workflow:
The system fits the existing library workflow and supports daily operations without disrupting routine activities.

Training and Support:
Training sessions will be conducted to ensure staff and users can use the system effectively, increasing user confidence and reducing resistance to change.

Conclusion:
The proposed system is operationally feasible because it meets user requirements, improves current processes, enhances efficiency, and fits well into the organization's existing workflow.

2.4 Additional Feasibility Considerations

Although not always required in project reports, the following factors are sometimes reviewed:

1. Schedule/Time Feasibility:
Evaluates whether the project can be completed within a specified period. It ensures that timelines are realistic and achievable based on available resources.

2. Legal Feasibility:
Ensures there are no legal issues related to data protection, software licensing, or regulatory requirements that might impede the project.

3. Risk Analysis:
Identifies potential risks (technical, financial, internal resistance) and prepares mitigation plans to minimize impact during implementation.

2.5 Summary and Final Assessment

A comprehensive feasibility study has been conducted for the Library Management System based on technical, economic, and operational aspects:

Technical: Available technology and expertise support system development.

Economic: Costs are justified by long- term benefits and savings.

Operational: System is user- friendly, fits organizational needs, and improves efficiency.

CHAPTER-3

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 INTRODUCTION TO SRS

A Software Requirements Specification (SRS) is a comprehensive document that outlines the functional and non-functional requirements of a software system, serving as a blueprint for development. It details the purpose, scope, target audience, and features of the software, ensuring all stakeholders—developers, testers, clients, and project managers—share a common understanding of the project. The SRS typically includes user requirements, system requirements, interface descriptions, constraints, and performance metrics. By providing a clear and structured framework, the SRS minimizes ambiguities, reduces development risks, and serves as a reference throughout the software lifecycle.

3.2 SOFTWARE REQUIREMENTS

Software Requirement for Server:

Web server	-	Apache (WAMPP)
Back End	-	PHP
Front End	-	HTML, CSS
Operating System	-	Windows 11 and above
IDE	-	Visual Studio
Database	-	MySQL

Software Requirement for Client's:

Connection	-	Having well-establish internet connection
Operating System	-	Windows 11
Web Browser	-	Chrome, Edge, etc.

3.3 HARDWARE REQUIREMENTS

Hardware Requirement for Client's:

Monitor/Display - 13" LCD monitor

Keyboard - Qwerty

Processor - Dual-Core 2Ghz or higher

Hard Disk - 32GB

Minimum Free Space - 500 MB free disk space

Minimum RAM - 2GB RAM

Hardware Requirement for Server:

Monitor/Display - 13" LCD monitor

Keyboard - Qwerty

Processor - Dual-Core 2Ghz or higher

Hard Disk - 32GB

Minimum Free Space - 500 MB free disk space Minimum RAM 2

CHAPTER-4

4. SYSTEM DESIN

4.1 PROCESS FLOW DIAGRAM

Process Flow Diagram and Flowchart is a diagram which uses geometric symbols and arrows to define relationships. It is a diagrammatic representation of the algorithm. The Process Flow Diagram of our application is shown below.

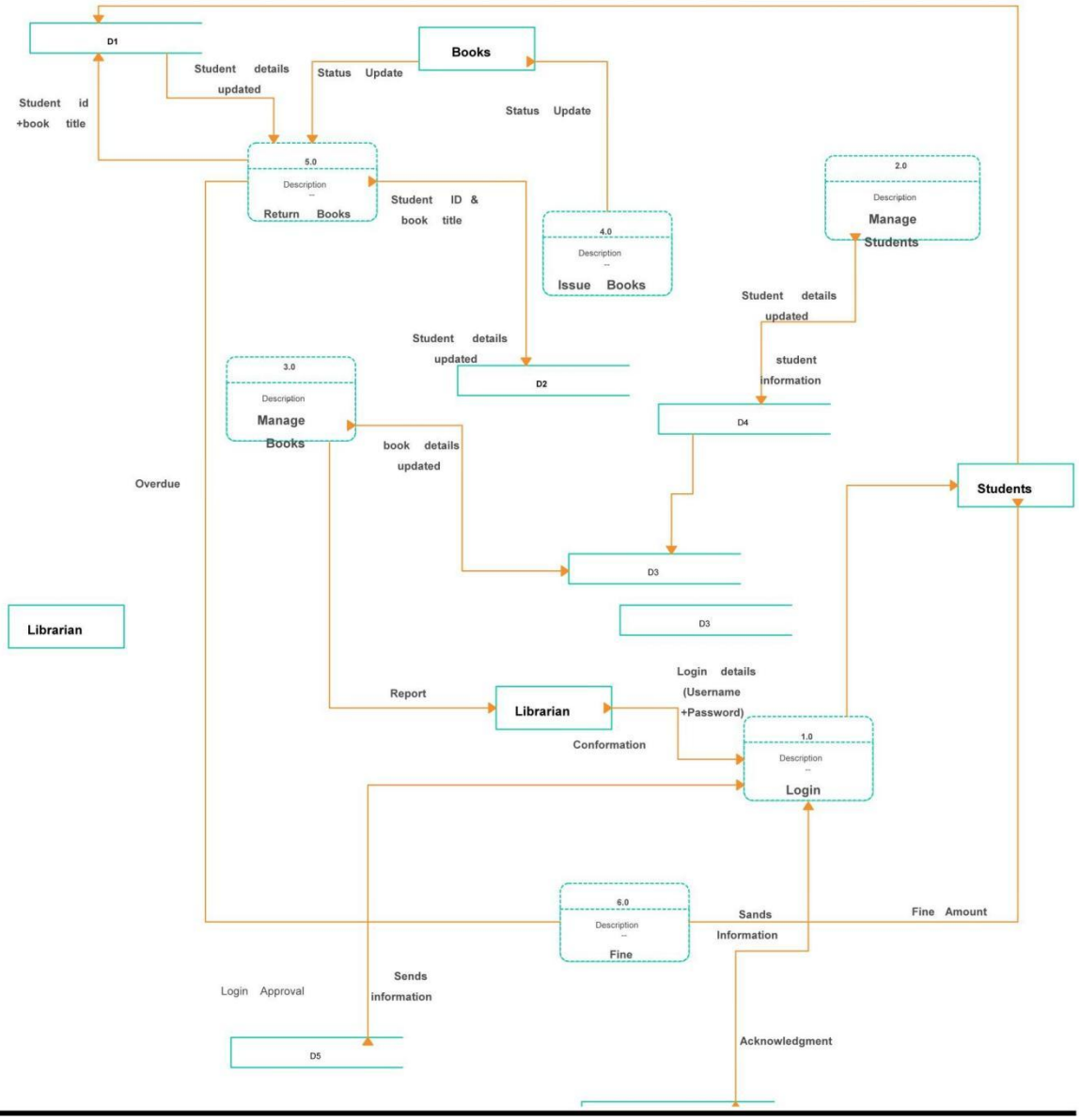


Figure 1 FLOW CHART

4.2 ER DIAGRAM

An ER (Entity-Relationship) diagram is a visual representation of the entities in a system and the relationships between them. It's commonly used in database design to map out the structure of the data and how different entities interact.

Here's a brief explanation of the components of an ER diagram:

Components of an ER Diagram:

(a) Entities: These are the objects or things in the system that have data stored about them. For example:

- Student
- Course
- Employee

Entities are usually represented by rectangles.

(b) Attributes: These are the data or properties that describe the entities. For example:

- A Student entity might have attributes like Student_ID, Name, Age.
- An attribute is typically represented by an oval.

(c) Relationships: These represent the interactions or associations between entities. For example:

- A Student "enrolls" in a Course.

Relatinships are typically represented by diamonds.



Figure 2 COMPONENTS OF AN ER DIAGRAM

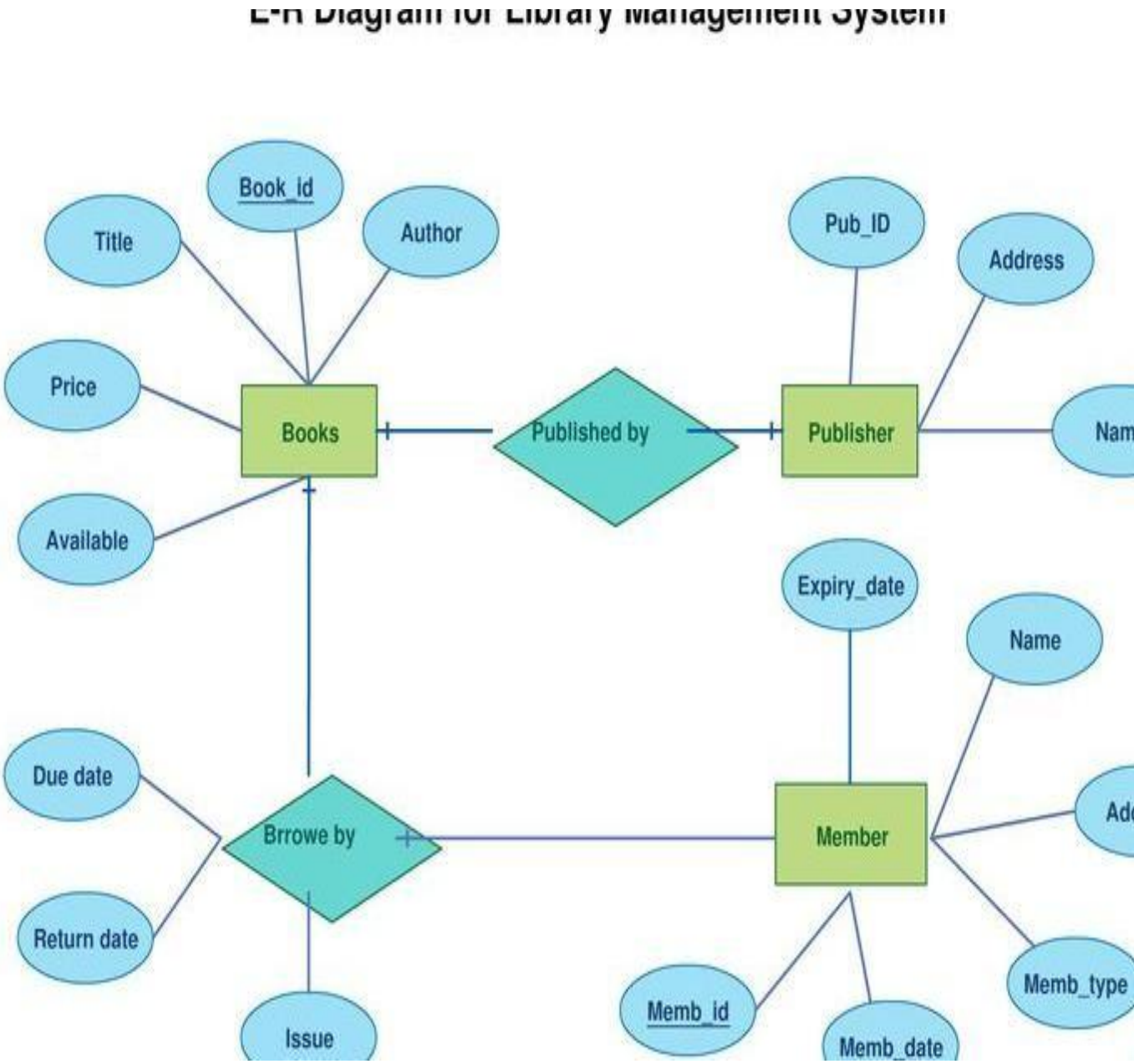


Figure 3 ER DIAGRAM

4.3 DATA FLOW DIAGRAM

A DFD (Data Flow Diagram) is a visual representation that illustrates how data moves within a system, showing how inputs are transformed into outputs through various processes. It is commonly used in software engineering and system analysis to model the flow of data within an application or business process.

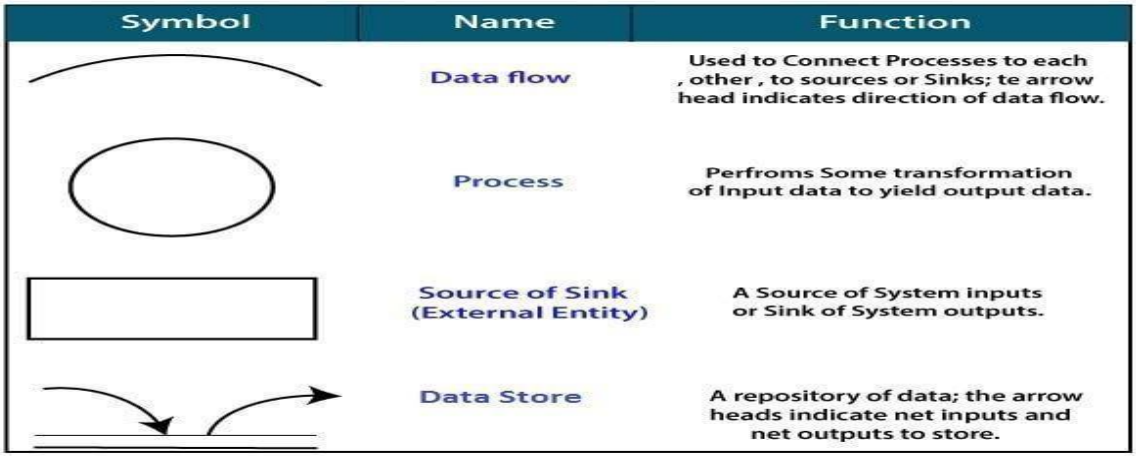
Key Components of a DFD:

(d) Processes: These represent activities or functions that transform input data into output. Processes are typically represented by circles or rounded rectangles.

(e) Data Stores: These are repositories where data is stored for later use. They are usually represented by open-ended rectangles.

(f) Data Flows: These represent the movement of data between processes, data stores, and external entities. Data flows are represented by arrows.

(g) External Entities: These are sources or destinations of data that are outside the system being modeled. They are typically represented by rectangles or squares.



Symbols for Data Flow Diagrams

CHAPTER-5

5. TOOLS AND TECHNOLOGY

5.1 DEVELOPMENT TOOLS

VISUAL-STUDIO-CODE:

Visual Studio Code (VS Code) is a popular, lightweight, and versatile code editor developed by Microsoft. It supports a wide range of programming languages and is equipped with features like syntax highlighting, intelligent code completion, debugging, and Git integration, making it a powerful tool for developers. VS Code's extensibility allows users to enhance its functionality by adding extensions for specific frameworks, languages, or tools, enabling customization for various development needs. Its user-friendly interface and cross-platform compatibility (Windows, macOS, and Linux) make it an ideal choice for developers of all levels, from beginners to professionals. Whether you're building websites, writing scripts, or developing software, VS Code provides a seamless and efficient coding experience. VS Code provided integrated terminal to run shell commands without leaving the editor and also Out-of-the-box support for languages like JavaScript, Python, C++, and many more.

WAMPP:

WAMP is an acronym that stands for **Windows, Apache, MySQL, and PHP**. It refers to a **software stack** used to create a local web server environment on a Windows computer for web development and testing purposes.

5.2 TECHNOLOGY USED

PHP:

PHP (Hypertext Preprocessor) is a widely-used, open-source server-side scripting language designed primarily for web development. It is embedded within HTML and is especially suited for creating dynamic and interactive web pages. PHP is known for its simplicity, flexibility, and extensive library of functions, making it a favorite among developers. It efficiently interacts with databases like MySQL, PostgreSQL, and others, enabling the development of robust web applications such as content management systems, e-commerce platforms, and social networking sites. With its broad community support, frequent updates, and compatibility with various operating systems, PHP remains a reliable and cost-effective solution for web development.

MYSQL-SERVER(phpMyAdmin):

MySQL, a powerful and widely-used open-source relational database management system, is often paired with PHP to create dynamic and data-driven web applications. It allows developers to efficiently store, retrieve, and manage large volumes of structured data. SQL (Structured Query Language) is used to interact with MySQL databases, providing commands to query and manipulate data. phpMyAdmin, a user-friendly web-based interface written in PHP, simplifies the management of MySQL databases. It allows developers to perform database tasks such as creating tables, running queries, and managing users without requiring extensive command-line knowledge. Together, MySQL and phpMyAdmin provide a seamless and effective environment for developing, testing, and managing web applications.

HTML:

HTML (HyperText Markup Language) is the standard language for creating and structuring content on the web. It provides the foundation for web pages by using elements and tags to define text, images, links, and other multimedia content. HTML is easy to learn and forms the backbone of web development, often working

alongside CSS for styling and JavaScript for interactivity. It is versatile and widely supported, enabling developers to build websites that are accessible across various devices and browsers. With advancements like HTML5, it supports modern web features such as multimedia playback, graphics rendering, and responsive design.

CSS:

CSS (Cascading Style Sheets) is a stylesheet language used to control the presentation and layout of web pages written in HTML. It allows developers to apply styles such as colors, fonts, spacing, and positioning to elements, creating visually appealing and organized websites. By separating content (HTML) from design (CSS), it enhances the maintainability and scalability of web projects. CSS supports responsive design, enabling web pages to adapt seamlessly to different screen sizes and devices. With features like animations, transitions, and grid layouts, CSS adds dynamic and interactive elements to web pages.

5.3 WEB-SERVER

APACHE:

Apache (or Apache HTTP Server) is a widely-used open-source web server software that serves content over the web. It is responsible for handling HTTP requests from clients (typically web browsers), processing those requests, and returning the appropriate content, such as HTML pages, images, or other files. Apache is highly configurable and can run on various operating systems, including Linux, Windows, and macOS.

As a web server, Apache is one of the most popular choices for hosting websites and web applications. It can be used to serve both static content (like HTML files) and dynamic content (such as PHP or CGI scripts). Apache also supports features like URL rewriting, access control, security configurations, and modules for extending its functionality, such as integrating with PHP or SSL for secure connections.

In summary, Apache is the server that hosts and delivers web content to users, playing a critical role in the infrastructure of most websites.

CHAPTER-6

SCREENSHOTS:

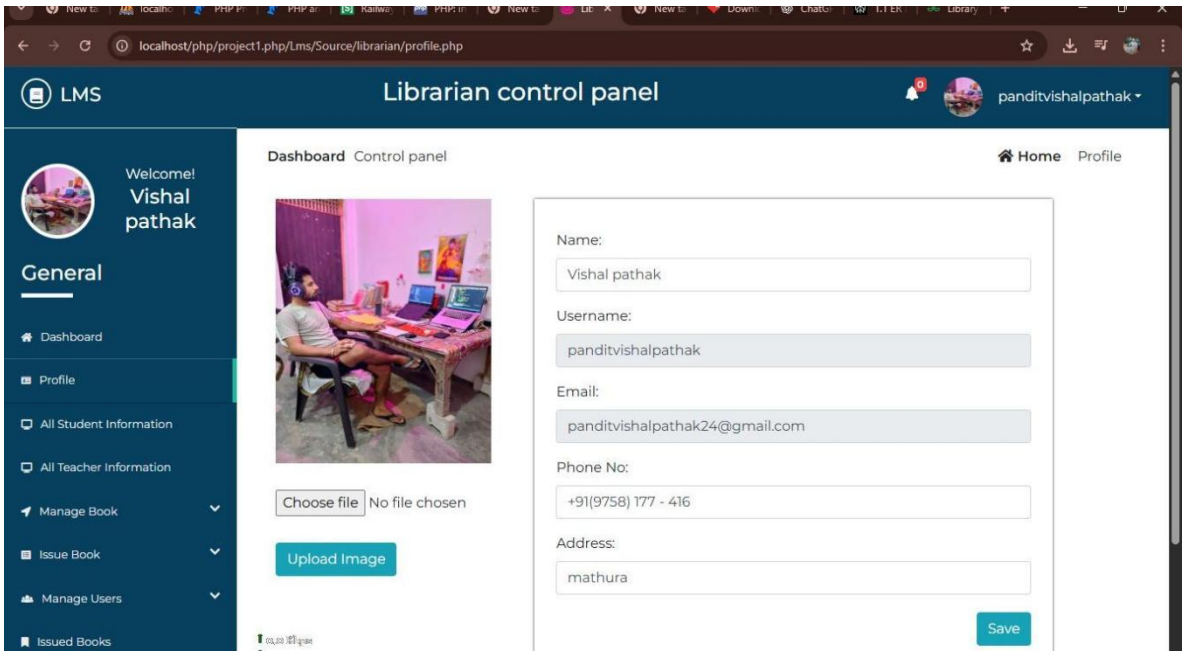


Figure 5 profile.php

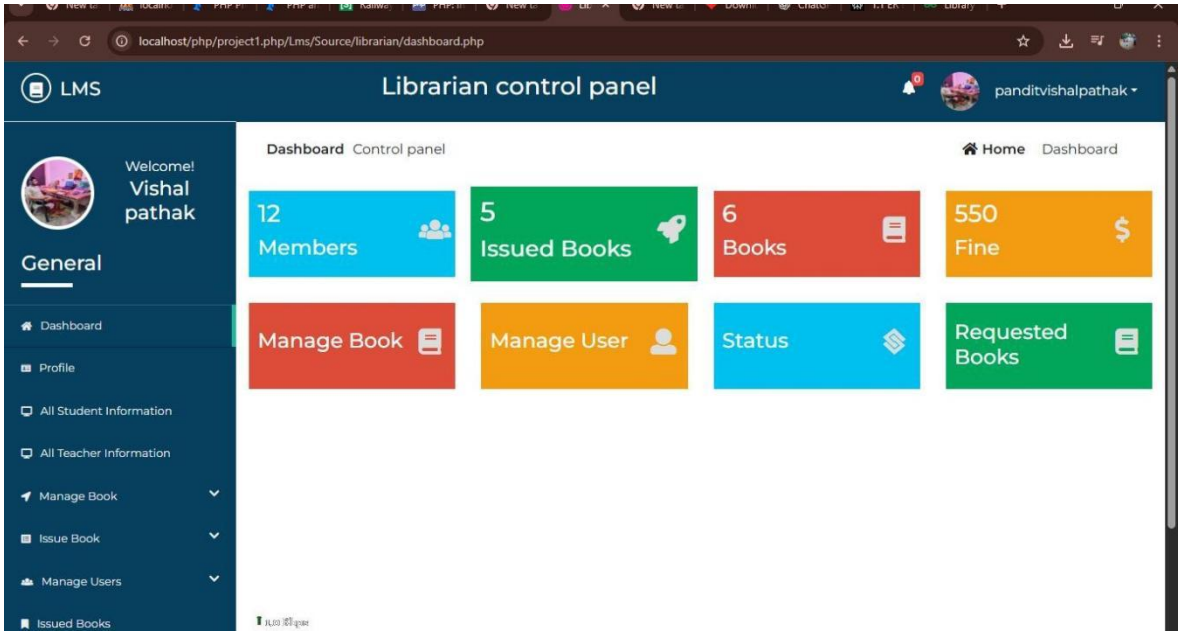


Figure 6 DASHBOARD

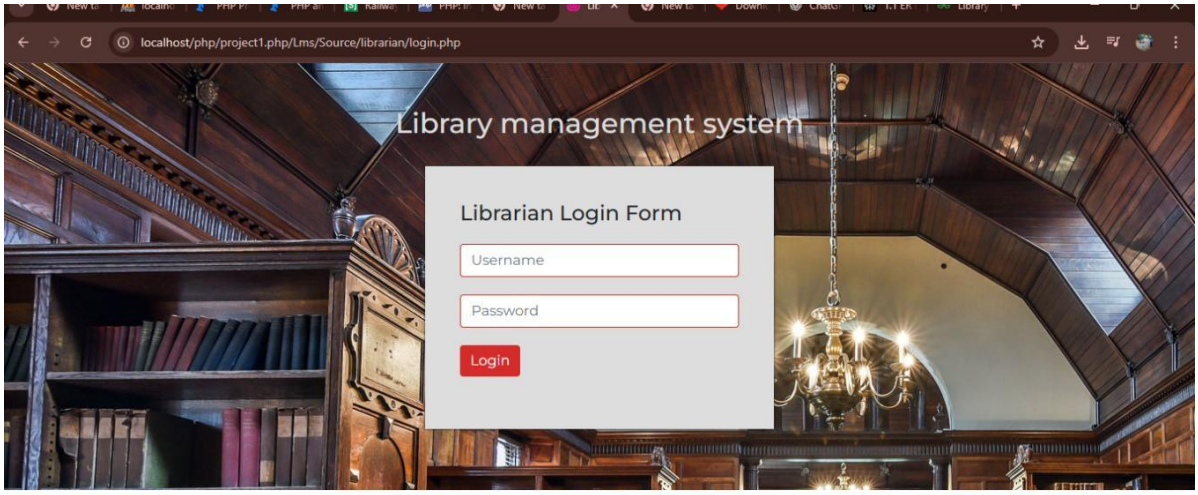


Figure 7 LOGIN PAGE

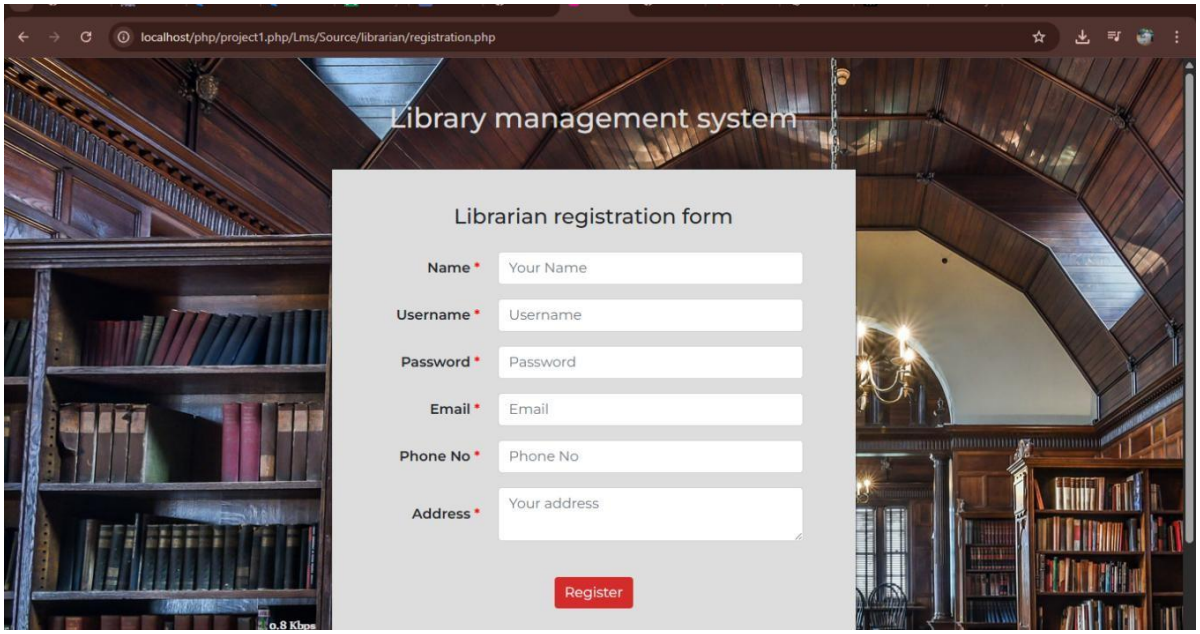


Figure 8 Registration Form

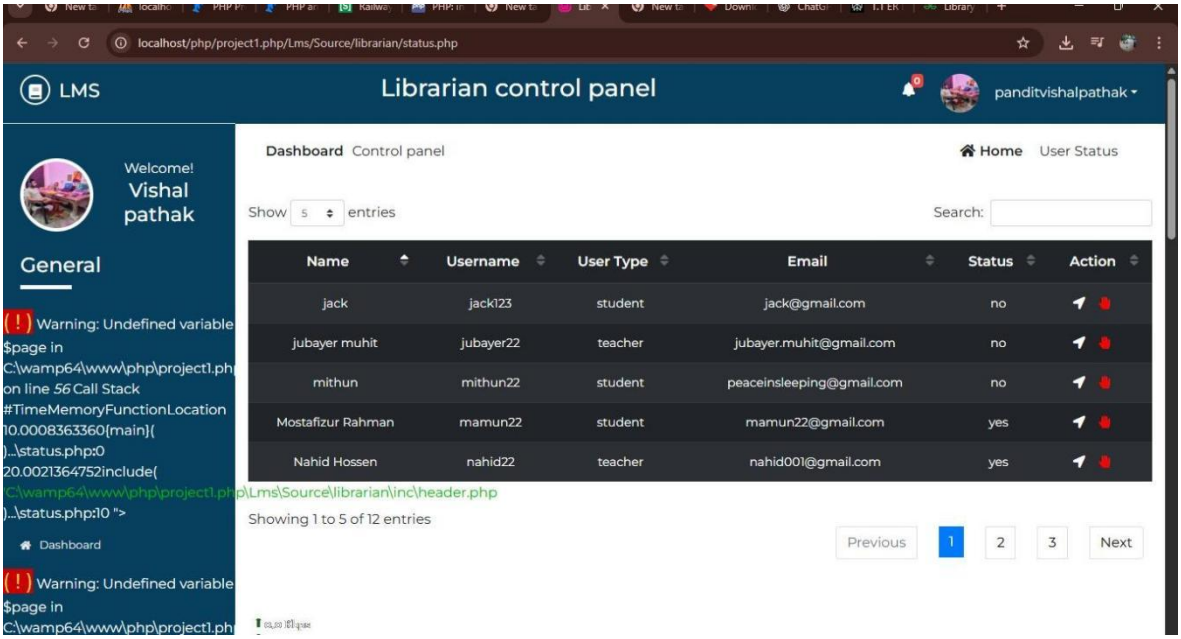


Figure 9 STATUS.PHP

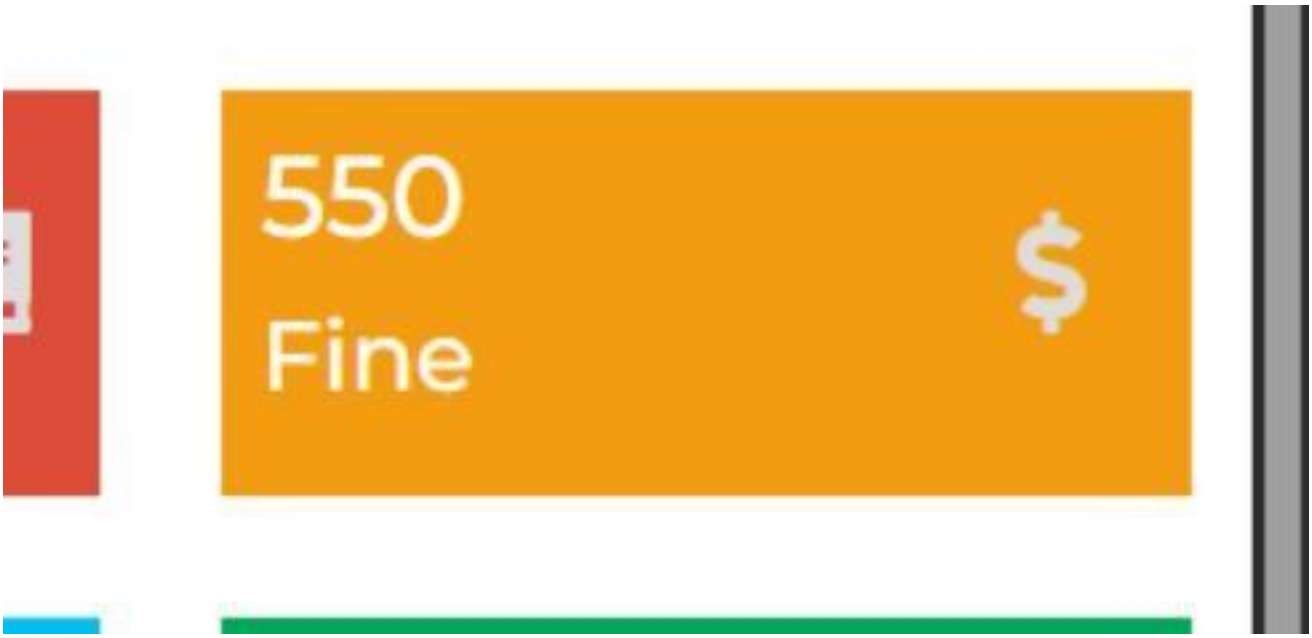


Figure 10 Fine.php

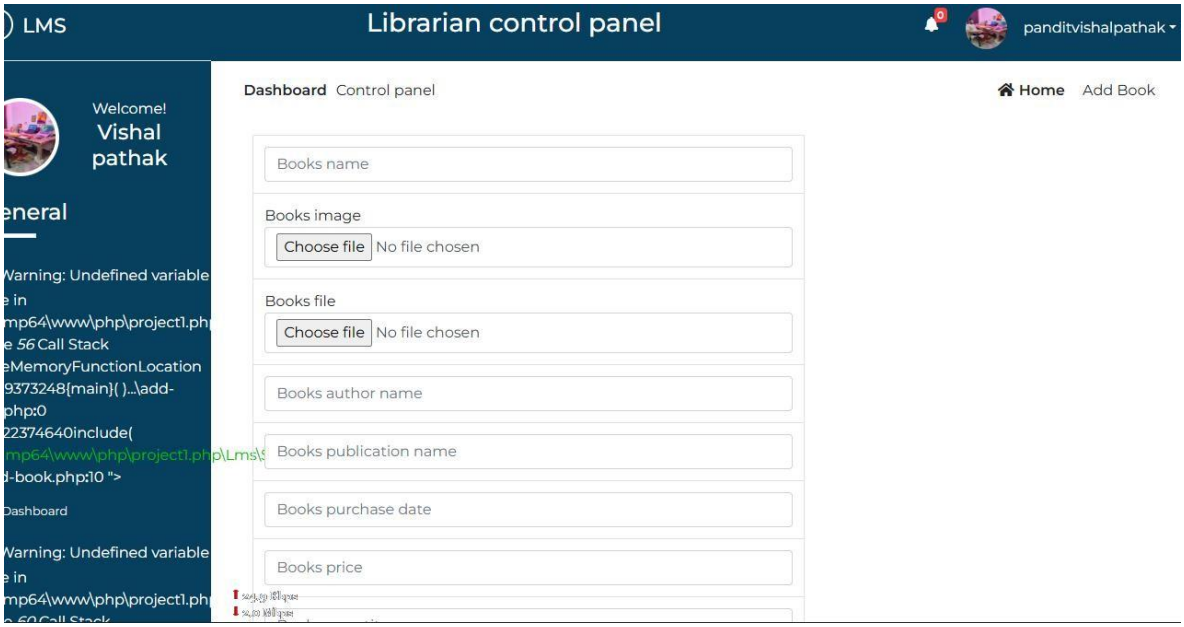


Figure 11 ADD BOOK

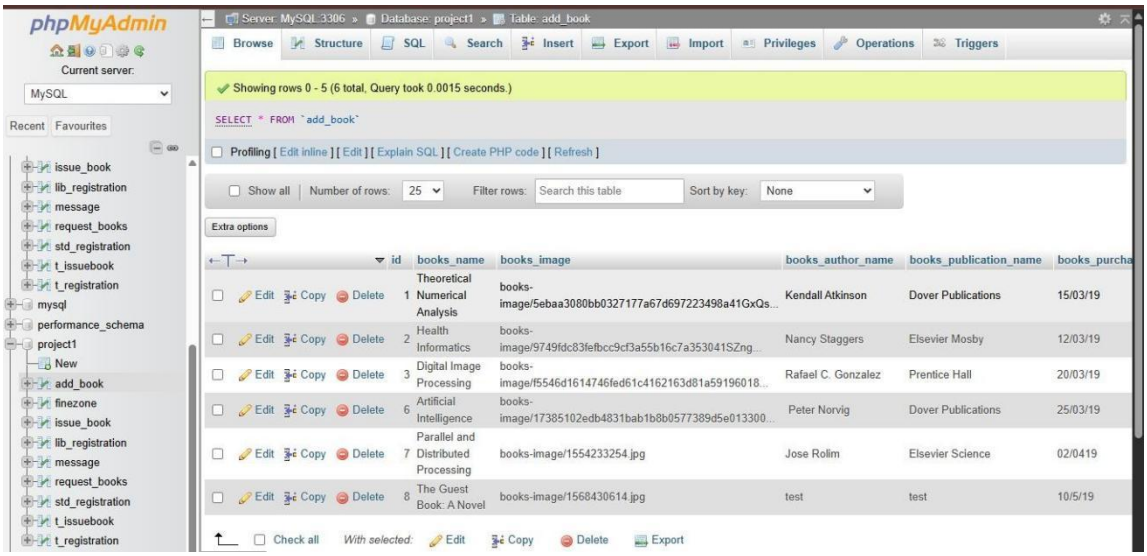


Figure 12 DATABASE 1

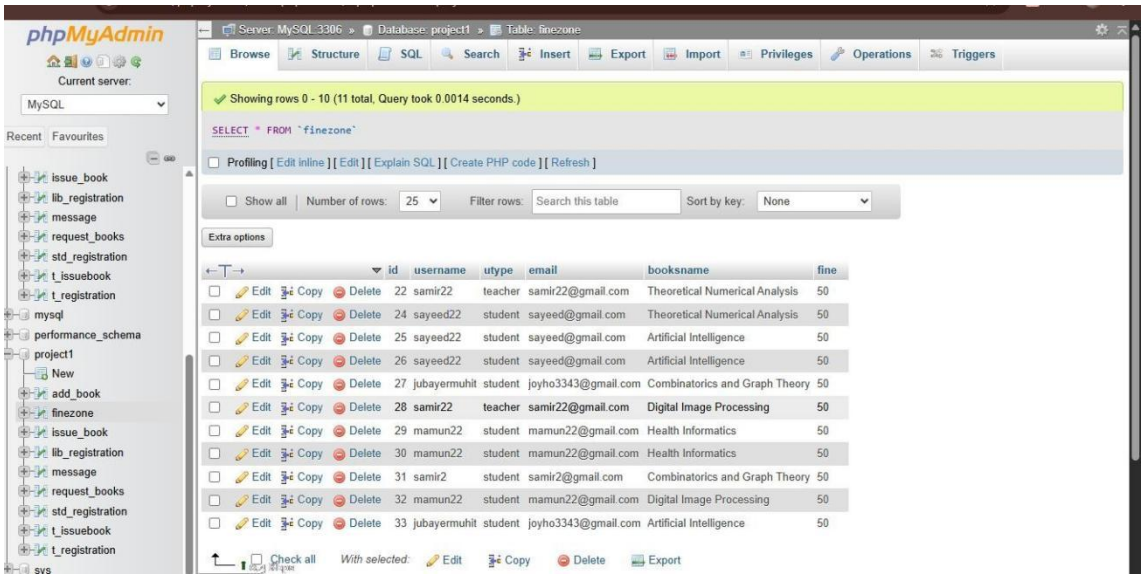


Figure 13 DATABASE 2

CHAPTER-7

7. CODING

config.php:

```
<?php
$SERVER =  "localhost";
$USERNAME = "root";
$PASSWORD =  "";
$DBNAME = "lms";
$conn = mysqli_connect($SERVER, $USERNAME, $PASSWORD, $DBNAME);
 if($conn)
{
    echo "connection ok";
}
else{
    echo "connection failed".mysqli_connect_error();
}
?>


<?php include("config.php");
?>
```

index.php:

```
<?php
   session_start();
   include 'inc/connection.php';
?>
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Library Management System</title>
   <link rel="stylesheet" href="inc/css/bootstrap.min.css">
   <link rel="stylesheet" href="inc/css/fontawesome-all.min.css">
   <link rel="stylesheet" href="inc/css/pro1.css">
   <link href="https://fonts.googleapis.com/css?family=Montserrat:400,500,600"
rel="stylesheet">
   <style>
      .login{
         background-image: url(inc/img/3.jpg);
         margin-bottom: 30px;
         padding: 50px;
         padding-bottom: 70px;
      }
      .reg-header h2{ color:
         #DDDDDD; z-
         index: 999999;
      }
      .login-body h4{
         margin-bottom: 20px;
      }
   </style>
</head>
<body>
   <div class="login registration">
      <div class="wrapper">
```

```
        <div class="reg-header text-center">
            <h2>Library management system</h2>
            <div class="gap-30"></div>
            <div class="gap-30"></div>
        </div>
        <div class="gap-30"></div>
        <div class="login-content">
            <div class="login-body">
                <h4>Librarian Login Form</h4>
                <form action="" method=”post">
                    <div class="mb-20">
                        <input type="text" name="username" class="form-control"
placeholder="Username"
required=""/>
                    </div>
                    <div class="mb-20">
                        <input type="password" name="password" class="form-control"
placeholder="Password" required=""/>
                    </div>
                    <div class="mb-20">
                        <input class="btn btn-info submit" type="submit" name="login"
value="Login">

                    </div>
                </form>
            </div>
            <?php
            if (isset($_POST["login"])) {
                $count=0;
                $res= mysqli_query($link, "select * from lib_registration where
username='$_POST[username]' && password= '$_POST[password]' ");
                $count = mysqli_num_rows($res);
                if ($count==0) {
                    ?>
                    <div class="alert alert-warning">
                        <strong style="color:#333">Invalid!</strong> <span style="color:
```

```
red;font-weight: bold; ">Username Or Password.</span>
                    </div>
                <?php
                }
                else{
                $_SESSION["username"] = $_POST["username"];
                ?>
                    <script type="text/javascript">
                        window.location="dashboard.php";
                    </script>
                    <?php
                }
            }
            ?>
          </div>
        </div>
      </div>
      <div class="footer text-center">
        <p>&copy; All rights reserved utter pompously</p>
      </div>

    <script src="inc/js/jquery-2.2.4.min.js"></script>
    <script src="inc/js/bootstrap.min.js"></script>
    <script src="inc/js/custom.js"></script>
    </body>
    </html>
```

registration.php:

```
<?php
   include 'inc/connection.php';
   include 'inc/function.php';
 ?>
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
```

```
    <title>Library Management System</title>
    <link rel="stylesheet" href="inc/css/bootstrap.min.css">
    <link rel="stylesheet" href="inc/css/fontawesome-all.min.css">
    <link rel="stylesheet" href="inc/css/pro1.css">
    <link href="https://fonts.googleapis.com/css?family=Montserrat:400,500,600"
rel="stylesheet">
    <style>
        .registration{
            background-image: url(inc/img/3.jpg);
            margin-bottom: 30px;
            padding: 50px;
            padding-bottom: 70px;
        }
        .reg-header
            h2{ color:
            #DDDDDD; z-
            index: 999999;
        }
        .login-body h4{
            margin-bottom: 20px;
        }
    </style>
</head>
<body>
    <div class="registration">
        <div class="reg-wrapper">
            <div class="reg-header text-center">
                <h2>Library management system</h2>
            </div>
            <div class="gap-40"></div>
            <div class="reg-body">
                <h4 style="text-align: center; margin-bottom: 25px;">Student registration
form</h4>

                <form action="" class="form-inline" method="post">
                <?php if(isset($s_msg)):?>
                    <span class="success"> <?php echo $s_msg; ?></span>
                <?php endif ?>
                <?php if(isset($error_m)):?>
```

 <?php echo $error_m; ?>

```
            <?php endif ?>
                <div class="form-group">
                  <label for="name" class="text-right">Name <span>*</span></label>
                  <input type="text" class="form-control custom" placeholder="Your Name"
name="name"/>
                </div>
                <div class="form-group">
                    <label for="username">Username <span>*</span></label>
                  <input type="text" class="form-control custom" placeholder="Username"
name="username" />
                </div>
                 <?php if(isset($error_ua)):?>
                 <span class="error"> <?php echo $error_ua; ?></span>
                  <?php endif ?>
                  <?php if(isset($error_uname)):?>
                 <span class="error"> <?php echo $error_uname; ?></span>
                  <?php endif ?>
                <div class="form-group">
                    <label for="password">Password <span>*</span></label>
                  <input type="password" class="form-control custom"
placeholder="Password" name="password"/>
                </div>
                <?php if(isset($error_pw)):?>
                 <span class="error"> <?php echo $error_pw; ?></span>
                  <?php endif ?>

                <div class="form-group">
                    <label for="email">Email <span>*</span></label>
                  <input type="text" class="form-control custom" placeholder="Gmail"
name="email"/>
                </div>
                <?php if(isset($e_msg)):?>
                <span class="error"><?php echo $e_msg; ?> </span>
                <?php endif ?>
                <?php if(isset($error_email)):?>
                <span class="error"><?php echo $error_email; ?> </span>
                <?php endif ?>
                <div class="form-group">
                    <label for="phone">Phone No <span>*</span></label>
```

```
                <input type="text" class="form-control custom" placeholder="Phone No"
name="phone"/>
            </div>
            <?php if(isset($error_phone)):?>
            <span class="error"><?php echo $error_phone; ?></span>
              <?php endif ?>
            <div class="form-group">
               <label for="sem">Select Semester <span>*</span></label>
               <select class="form-control custom" name="sem">
                  <option>1th</option>
                  <option>2nd</option>
                  <option>3rd</option>
                  <option>4th</option>
                  <option>5th</option>
                  <option>6th</option>
                  <option>7th</option>
                  <option>8th</option>
               </select>
            </div>
            <div class="form-group">
               <label for="dept">Department <span>*</span></label>
               <select class="form-control custom" name="dept">
                  <option>CSE</option>
                  <option>EEE</option>
                  <option>ECE</option>
                  <option>BBA</option>
                  <option>Others</option>
               </select>
            </div>
            <div class="form-group">
                <label for="session">Session <span>*</span></label>
               <input type="text" class="form-control custom" placeholder="14/15"
name="session"/>
            </div>
            <div class="form-group">
                <label for="regno">Registration No <span>*</span></label>
               <input type="text" class="form-control custom" placeholder="Registration
No" name="regno"/>
            </div>
```

```
                <?php if(isset($error_reg)):?>
                <span class="error"><?php echo $error_reg; ?></span>
                  <?php endif ?>
                <div class="form-group">
                    <label for="address">Address <span>*</span></label>
                    <textarea name="address" id="address"  class="form-control custom"
placeholder="Your address"></textarea>
                </div>
                <div class="submit">
                    <input type="submit" value="Register" name="submit" class="btn change
text-center">
                </div>
            </form>
        </div>
    </div>
  </div>

  <div class="footer text-center">
    <p>&copy; All rights reserved utter pompously</p>
  </div>

  <script  src="inc/js/jquery-2.2.4.min.js"></script>
  <script src="inc/js/bootstrap.min.js"></script>
  <script src="inc/js/custom.js"></script>
</body>
</html>
```

login.php:

```
<?php
  session_start();
  include 'inc/connection.php';
 ?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Library Management System</title>
  <link rel="stylesheet" href="inc/css/bootstrap.min.css">
```

```
    <link rel="stylesheet" href="inc/css/fontawesome-all.min.css">
    <link rel="stylesheet" href="inc/css/pro1.css">
    <link href="https://fonts.googleapis.com/css?family=Montserrat:400,500,600"
rel="stylesheet">
    <style>
        .login{
            background-image: url(inc/img/3.jpg);
            margin-bottom: 30px;
            padding: 50px;
            padding-bottom: 70px;
        }
        .reg-header h2{ color:
            #DDDDDD; z-
            index: 999999;
        }
        .login-body h4{
            margin-bottom: 20px;
        }
    </style>
</head>
<body>

    <div class="login registration">
        <div class="wrapper">
            <div class="reg-header text-center">
                <h2>Library management system</h2>
                <div class="gap-40"></div>
            </div>
            <div class="gap-30"></div>
            <div class="login-content">
                <div class="login-body">
                    <h4>User Login Form</h4>
                    <form action="" method="post">
                        <div class="mb-20">
                            <input type="text" name="username" class="form-control"
```

```
placeholder="Username" required=""/>
                </div>
                <div class="mb-20">
                  <input type="password" name="password" class="form-control"
placeholder="Password" required=""/>
                </div>
                <div class="mb-20">
                  <input class="btn btn-info submit" type="submit" name="login"
value="Login">
                  <a class="reset_pass" href="lost-password.php">Lost your password?</a>
                </div>
              </form>
            </div>
            <div class="login-footer text-center">
              <div class="separator">
                <p class="change_link">New to site?
                  <a href="registration.php" class="text-right"> Create Account </a>
                </p>
              </div>
            </div>
            <?php
              if (isset($_POST["login"])) {
                $count=0;
                $res= mysqli_query($link, "select * from std_registration where
username='$_POST[username]' && password= '$_POST[password]' && status='yes' &&
verified='yes' ");
                $count = mysqli_num_rows($res);
                if ($count==0) {
                  ?>
                    <div class="alert alert-warning">
                    <strong style="color:#333">Invalid!</strong> <span style="color:
red;font-weight: bold; ">Username Or Password.</span>
                    </div>
                  <?php
                }
                else{
```

```
                    $_SESSION["student"] = $_POST["username"];
                    ?>
                    <script type="text/javascript">
                       window.location="my-issued-books.php";
                    </script>
                    <?php
                  }
               }
            ?>
         </div>
      </div>
   </div>

   <div class="footer text-center">
      <p>&copy; All rights reserved utter pompously</p>
   </div>

   <script src="inc/js/jquery-2.2.4.min.js"></script>
   <script src="inc/js/bootstrap.min.js"></script>
   <script src="inc/js/custom.js"></script>
</body>
</html>
```

profile.php:

```
<?php
    session_start();
   if (!isset($_SESSION["student"])) {
      ?>
         <script type="text/javascript">
            window.location="login.php";
         </script>
      <?php
   }
   include 'inc/header.php';
   include 'inc/connection.php';
```

```
?>
  <!--dashboard area-->
  <div class="dashboard-content">
    <div class="dashboard-header">
      <div class="container">
        <div class="row">
          <div class="col-md-6">
            <div class="left">
              <p><span>dashboard</span>User panel</p>
            </div>
          </div>
          <div class="col-md-6">
            <div class="right text-right">
              <a href="dashboard.php"><i class="fas fa-home"></i>home</a>
              <span class="disabled">profile</span>
            </div>
          </div>
        </div>
        <div class="profile-content">
          <div class="row">
            <div class="col-md-3">
              <div class="photo">
                <?php
                  $res = mysqli_query($link, "select * from std_registration where
username='".$_SESSION['student']."'");
                  while ($row = mysqli_fetch_array($res)){
                    ?><img src="<?php echo $row["photo"]; ?> " height=""
width="" alt="something wrong"></a> <?php
                  }

                ?>
              </div>
              <div class="uploadPhoto">
                <div class="gap-30"></div>
                <form action="" method="post" enctype="multipart/form-data">
```

```
                        <input type="file" name="image" class="modal-mt" id="image">
                        <div class="gap-30"></div>
                        <input type="submit" class="modal-mt btn btn-info"
value="Upload Image" name="submit">
                    </form>
                </div>
                <?php
                    if (isset($_POST["submit"])) {
                        $image_name=$_FILES['image']['name'];
                        $temp = explode(".", $image_name);
                        $newfilename = round(microtime(true)) . '.' . end($temp);
                        $imagepath="upload/".$newfilename;
                        move_uploaded_file($_FILES["image"]["tmp_name"],$imagepath)
;
                        mysqli_query($link, "update std_registration set
photo='".$imagepath."' where username='".$_SESSION['student']."'");
                        ?>
                            <script type="text/javascript">
                                window.location="profile.php";
                            </script>
                        <?php
                    }
                ?>
            </div>
            <div class="col-md-9">
                <div class="details">
                    <?php
                        $res5 = mysqli_query($link, "select * from std_registration
where username='$_SESSION[student]' ");
                        while($row5 = mysqli_fetch_array($res5)){
                            $regno      = $row5['regno'];
                            $username  = $row5['username'];
                            $name      = $row5['name'];
                            $sem      = $row5['sem'];
                            $session      = $row5['session'];
                            $dept      = $row5['dept'];
```

```
                        $email     = $row5['email'];
                        $phone     = $row5['phone'];
                        $address     = $row5['address'];
                        $utype     = $row5['utype'];
                    }
                ?>
            <form method="post">
                <div class="form-group details-control">
                    <label for="regno" class="text-right">Reg No:</label>
                    <input type="text" class="form-control custom"  name="regno"
value="<?php echo $regno; ?>" disabled />
                </div>
                <div class="form-group details-control">
                     <label for="username">Username:</label>
                    <input type="text" class="form-control
custom"  name="username" value="<?php echo $username; ?>" disabled />
                </div>
                <div class="form-group details-control">
                    <label for="name" class="text-right">Name:</label>
                    <input type="text" class="form-control custom"  name="name"
value="<?php echo $name; ?>" />
                </div>

                <div class="form-group details-control">
                    <label for="sem" class="text-right">Semester:</label>
                    <input type="text" class="form-control custom"  name="sem"
value="<?php echo $sem; ?>" />
                </div>
                <div class="form-group details-control">
                    <label for="session" class="text-right">Session:</label>
                    <input type="text" class="form-control
custom"  name="session" value="<?php echo $session; ?>" />
                </div>
                <div class="form-group details-control">
                    <label for="dept" class="text-right">Departemt:</label>
                    <input type="text" class="form-control custom"  name="dept"
```

```
value="<?php echo $dept; ?>" />
                        </div>

                        <div class="form-group details-control">
                            <label for="email">Email:</label>
                           <input type="text" class="form-control custom"  name="email"
value="<?php echo $email; ?>" disabled/>
                        </div>
                        <div class="form-group details-control">
                            <label for="phone">Phone No:</label>
                           <input type="text" class="form-control custom"  name="phone"
value="<?php echo $phone; ?>" />
                        </div>
                        <div class="form-group details-control">
                           <label for="address">Address:</label>
                            <input type="text" class="form-control
custom"  name="address" value="<?php echo $address; ?>" />
                        </div>
                        <div class="form-group details-control">
                           <label for="utype">User Type:</label>
                            <input type="text" class="form-control custom"  name="utype"
value="<?php echo $utype; ?>"  disabled="" />
                        </div>
                        <div class="text-right mt-20">
                           <input type="submit" value="Save" class="btn btn-info"
name="update">
                        </div>
                    <?php
                    ?>
                    </form>
                </div>
                <?php
                  if
                     (isset($_POST["update"])){ mysqli_query($lin
                     k, "update std_registration set
                     name='$_POST[name]',
```

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e
=
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e
]
'
,

```
                        address='$_POST[address]'
                        where username='$_SESSION[student]'");
                        ?>
                            <script type="text/javascript">
                                window.location="profile.php";
                            </script>
                        <?php
                    }
                ?>
            </div>
        </div>
    </div>
</div>
</div>
<?php
    include 'inc/footer.php';
?>
```

notification.php:

Book.php:

```
<?php
    session_start();
   if (!isset($_SESSION["student"])) {
      ?><?php
     session_start();
      if (!isset($_SESSION["student"])) {
         ?>
            <script type="text/javascript">
                window.location="login.php";
            </script>
         <?php
      }
      include 'inc/header.php';
```

```
        include 'inc/connection.php';
        mysqli_query($link,"update message set read1='y' where
rusername='$_SESSION[student]'");

    ?>
      <!--dashboard area-->
   <div class="dashboard-content">
      <div class="dashboard-header">
         <div class="container">
            <div class="row">
               <div class="col-md-6">
                  <div class="left">
                     <p><span>dashboard</span>User panel</p>
                  </div>
               </div>
               <div class="col-md-6">
                  <div class="right text-right">
                     <a href="dashboard.php"><i class="fas fa-home"></i>home</a>
                     <span class="disabled">notifications</span>
                  </div>
               </div>
               <div class="col-md-12">
                  <table class="table table-bordered text-center table-striped">
                     <tr>
                        <th>Librarian name</th>
                        <th>Title</th>
                        <th>Message</th>
                        <th>Time</th>
                     </tr>
                     <?php
                          $res=mysqli_query($link,"select * from message where
rusername='$_SESSION[student]' order by id desc");

                          while ($row = mysqli_fetch_array($res)){
                             $res1=mysqli_query($link,"select * from lib_registration where
username='$row[susername]'");
                             while ($row1 = mysqli_fetch_array($res1)){
                                $name = $row1["name"];
                             }
```

```
                        echo "<tr>";
                        echo "<td>"; echo $name; echo "</td>";
                        echo "<td>"; echo $row["title"]; echo "</td>";
                        echo "<td>"; echo $row["msg"]; echo "</td>";
                        echo "<td>"; echo $row["time"]; echo "</td>";
                        echo "</tr>";
                    }
                ?>
            </table>
        </div>
    </div>
</div>
</div>
</div>

<?php
    include 'inc/footer.php';
?>


    <script type="text/javascript">
        window.location="login.php";
    </script>
<?php
}
$page = 'books';
include 'inc/header.php';
include 'inc/connection.php';
?>
<!--dashboard area-->
<div class="dashboard-content">
    <div class="dashboard-header">
        <div class="container">
            <div class="row">
                <div class="col-md-6">
                    <div class="left">
                        <p><span>dashboard</span>User panel</p>
                    </div>
                </div>
                <div class="col-md-6">
                    <div class="right text-right">
```

```
                    <a href="dashboard.php"><i class="fas fa-home"></i>home</a>
                    <span class="disabled">books</span>
                </div>
            </div>
        </div>
        <div class="books">
            <form action="" method="post" name="form1">
                <table class="table ">
                    <tr>
                        <td>
                            <input type="text" name="search" class="form-control" placeholder="Enter
book name">
                        </td>
                        <td>
                            <input type="submit" name="submit1" class="btn btn-info" value="Search
Book">
                        </td>
                    </tr>
                </table>
            </form>
            <?php
                if (isset($_POST["submit1"])) {
                    $i=0;
                    $res = mysqli_query($link,"select * from add_book where books_name
like('$_POST[search]%')");
                    echo "<table class='table control-books'>";
                    echo "<tr>";
                    while ($row = mysqli_fetch_array($res)){
                        $i=$i+1;
                        echo "<td>";
                        ?> <a href="../<?php echo $row["books_file"]; ?>" target="_blank"><img
src="../../<?php echo $row["books_image"]; ?> " alt=""></a> <?php
                        echo "</br>";
                        echo "</br>";
                        echo "<b>".$row["books_name"]; "</b>";
                        echo "</br>";
                        echo "<b>".
                        "Available: ".$row["books_availability"]; "</b>";
                        echo "</td>";

                        if ($i>=4)
                            { echo
                            "</tr>"; echo
                            "<tr>";
                            $i=0;
                        }
```

```
                    }
                    echo "</tr>";
                    echo "</table>";
                    }
                    else{
                        $i=0;
                        $res = mysqli_query($link,"select * from add_book where books_availability>0");
                        echo "<table id='dtBasicExample' class='table control-books'>";
                        echo "<tr>";
                        while ($row = mysqli_fetch_array($res)){
                            $i=$i+1;
                            echo "<td>";
                            ?> <a href="../../<?php echo $row["books_file"]; ?>" target="_blank"><img
src="../../<?php echo $row["books_image"]; ?> " alt=""></a> <?php
                            echo "</br>";
                            echo "</br>";
                            echo "<b>".$row["books_name"]; "</b>";
                            echo "</br>";
                            echo "<b>".
                            "Available: ".$row["books_availability"]; "</b>";
                            echo "</td>";

                            if ($i>=4)
                                { echo
                                "</tr>"; echo
                                "<tr>";
                                $i=0;
                            }

                        }
                        echo "</tr>";
                        echo "</table>";
                        }
                ?>
              </div>
            </div>
          </div>
      </div>
      <?php
          include 'inc/footer.php';
       ?>
    <script>
      $(document).ready(function () {
      $('#dtBasicExample').DataTable();
      $('.dataTables_length').addClass('bs-select');
      });
```

```
</script>
```

verification.php:

```
<!doctype html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport"
        content="width=device-width, user-scalable=no, initial-scale=1.0, maximum-scale=1.0,
minimum-scale=1.0">
   <meta http-equiv="X-UA-Compatible" content="ie=edge">
   <title>Thank you for registration</title>
   <link rel="stylesheet" href="inc/css/bootstrap.min.css">
   <style>
      body{
         background: lightblue;
         padding: 100px;
      }
      .thankyou-
         content{ width:
         750px; margin: 0
         auto;
      }
     p{
        margin-top: 30px;
     }
   </style>
</head>
<body>
   <div class="thankyou-content text-center">
      <h4>Your Email is verified..</h4>
      <h4>Now your account is under maintenance. After approve your account we will send
you a email.</h4>
      <p><b>Thank you...</b></p>
   </div>

   <script src="inc/js/jquery-2.2.4.min.js"></script>
   <script src="inc/js/bootstrap.min.js"></script>
</body>
</html>
```

CHAPTER-8

8. TESTING

8.1 WHAT IS TESTING

Testing is a critical phase in the software development lifecycle that involves evaluating and validating a project to ensure it meets the required specifications and functions correctly. The primary goal of testing is to identify bugs, errors, or discrepancies in the system and verify that the software performs as expected under various conditions. For the Railway Reservation Management System, testing ensures that the application is reliable, efficient, and user-friendly.

8.2 TYPES OF TESTING

(a) Unit Testing: Unit testing is a method of testing individual units or components of a software application. It is typically done by developers and is used to ensure that the individual units of the software are working as intended. Unit tests are usually automated and are designed to test specific parts of the code, such as a particular function or method. Unit testing is done at the lowest level of the software development process, where individual units of code are tested in isolation.

(b) Integration Testing: Integration testing is a method of testing how different units or components of a software application interact with each other. It is used to identify and resolve any issues that may arise when different units of the software are combined. Integration testing is typically done after unit testing and before functional testing, and is used to verify that the different units of the software work together as intended.

(c) Regression Testing: Regression testing is a method of testing that is used to ensure that changes made to the software do not introduce new bugs or cause existing functionality to break. It is typically done after changes have been made to

the code, such as bug fixes or new features, and is used to verify that the software still works as intended.

(d) System Testing: System Testing is carried out on the whole system in the context of either system requirement specifications or functional requirement specifications or in the context of both. The software is tested such that it works fine for the different operating systems.

(e) Performance Testing: It is designed to test the run-time performance of software within the context of an integrated system. It is used to test the speed and effectiveness of the program. It is also called load testing. In it we check, what is the performance of the system in the given load.

(f) Black box Testing: Black Box Testing is a software testing method where the tester focuses on testing the functionality of the application without any knowledge of the internal workings or code structure.

(g) White box Testing: White Box Testing (also known as clear-box testing or structural testing) is a software testing method where the tester has full knowledge of the internal workings of the application, including the code, design, and architecture.

(i) Alpha Testing: Alpha testing is the initial testing phase where the software is tested by the internal development team or QA team. The goal is to identify bugs and issues before releasing the software to external users. It is performed in a controlled environment within the development organization.

(j) Beta Testing: Beta testing is conducted by a select group of external users (beta testers) before the software is released to the public. Beta testers provide feedback on the software's performance, usability, and potential issues. It helps identify any last-minute bugs or improvements.

8.3 TEST CASES OF THE PROJECT

TEST CASE 1: USER REGISTRATION

(a) Scenario: User wants to create a new account.

(b) Steps:

- User clicks on the "Signup" option.
- Enters valid details in the registration form.
- Submits the form.

(c) Expected Result: User account is successfully created, and the user is redirected to the main menu.

TEST CASE 2: USER LOGIN

(a) Scenario: User wants to log into their account.

(b) Steps:

- User clicks on the "Login" option.
- Enters valid credentials.
- Submits the form.

(c) Expected Result: User is successfully logged in and directed to the main menu.

TEST CASE 3: TRAIN SEARCH

(a) Scenario: User wants to search for available trains.

(b) Steps:

- User selects "Find Trains" from the main menu.
- Enters valid source, destination, and travel date.
- Submits the search request.

(c) Expected Result: System displays a list of available trains matching the criteria.

TEST CASE 4: SEAT RESERVATION

(a) Scenario: User wants to reserve seats on a selected train.

(b) Steps:

- User selects a train from the available options.
- Chooses seats and enters valid passenger details.
- Submits the reservation request.

(c) Expected Result: System confirms successful reservation and initiates payment.

TEST CASE 5: BOOKING HISTORY

(a) Scenario: User wants to view their booking history.

(b) Steps:

- User selects "Booking History" from the main menu.

(c) Expected Result: System displays a list of past and upcoming bookings for the user.

TEST CASE 6: LOGOUT

(a) Scenario: User wants to log out of their account.

(b) Steps:

- User selects "Logout" from the main menu.

(c) Expected Result: User session is terminated, and the user is redirected to the login page.

TEST CASE 7: INVALID LOGIN ATTEMPT

(a) Scenario: User tries to log in with incorrect credentials.

(b) Steps:

- User enters invalid username and password.
- Submits the login form.

(c) Expected Result: System displays an error message and prompts the user to re-enter valid credentials.

CHAPTER-9

9. CONCLUSION

In conclusion, the **Library Management System** developed as part of this project has successfully addressed the major challenges that exist in traditional manual library operations. Earlier, librarians had to maintain books, documents, and member records manually using registers, which was time- consuming, error- prone, and inefficient. The automated system replaces these outdated procedures with an organized, computerized process that improves reliability and enhances the overall functioning of the library.

The primary objectives of the project—including efficient book cataloging, member record management, issue and return handling, and quick retrieval of information—have been achieved through a centralized and user- friendly platform. Using this system, librarians can easily input, update, and search data, which results in significant time savings and minimized human errors. Furthermore, users are able to check book availability and see their borrowing details more conveniently, leading to a better user experience.

The system enhances operational efficiency by providing real- time updates on book status and transaction history. Overdue tracking and fine calculation features help enforce library policies effectively. The implementation of this system brings a structured approach to library management, ensuring accurate record- keeping and improved accessibility of information for both staff and patrons.

Overall, the Library Management System represents a meaningful technological improvement over manual processes. It offers a modern solution that supports efficient data handling, reduces workload, and makes daily library operations more systematic and reliable. With its potential for scalability and future enhancements—such as online access, integration with other educational systems, and advanced reporting—the system has laid a strong foundation for continued advancement in library services.

Thus, this project not only achieves its intended goals but also contributes to improving the quality of library management in educational and public settings in the digital age.

Here are some **references** you can include in your project report for the **Library Management System**. These cover books, research articles, and web sources that you may have referred to while developing or documenting your project:

Books

Elias Awath, *System Analysis and Design*, Tata McGraw- Hill Publication, Sixth Edition – useful for understanding system planning and design concepts.

S. Ramachandran, *Computer Aided Design*, Air Walk Publication, Third Edition – referenced for design principles in system projects.

Richard Fairley, *Software Engineering Concepts*, Tata McGraw- Hill Publication, Second Edition – covers foundational software engineering concepts used in system development.

Journal Articles & Research Papers

Jannathul Firthous et al., "Library Management System," *International Scientific Journal of Engineering and Management*, 2024 – discusses modern LMS approaches and applications.

K. Sugashini, Santhosh S, Yokeshwar P, Ghuru K., "Library Management System," *International Journal for Research in Applied Science and Engineering Technology*, 2023 – provides insights on LMS automation and efficiency.

Pawan Bhati, Komal, Prince & Stuti Saxena, *Modern Library Management System: A Web- Based Framework*, *International Journal of Research – GRANTHAALAYAH*, 2024 – explores resource automation and user oversight in LMS.

Sudarasan M., Thirshala Nandhini A., Kannan M., Sathiya T., *Library Management System*, *International Journal of Innovative Research in Engineering*, 2023 – article overview on LMS automation and workflow.

1.

Web Resources

Library Management System Project – GeeksforGeeks — provides software engineering details like requirements and implementation steps.

Library Management System Final Report Example — documentation template reference including bibliography and development structure.

Library Management Project Report Sample — example report structure used for guidance.

Reference Works (General Library Science)

Encyclopedia of Library and Information Sciences — a comprehensive reference for library theory and information management.

S.R. Ranganathan, *The Five Laws of Library Science* — foundational principles of library operations and service

Sample Reference List (APA Format)

Awath, E. (2003). *System Analysis and Design* (6th ed.). Tata McGraw- Hill Publication.

Fairley, R. (1997). *Software Engineering Concepts* (2nd ed.). Tata McGraw- Hill Publication.

Pressman, R. S. *Software Engineering: A Practitioner's Approach*. Tata McGraw- Hill Publishing.

Firthous, J., A., I., & Kumar, P. P. (2024). *Library Management System*. *International Scientific Journal of Engineering and Management*.

Bhati, P., Komal, Prince, & Saxena, S. (2024). *Modern Library Management System*. *International Journal of Research - GRANTHAALAYAH*.

Library Management System Project. (n.d.).

Encyclopedia of Library and Information Sciences. (n.d.).